

Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3410**

Title: **COMPOSITE BEAMS SECTIONS**

Revision: **1.0**

Date: **06/03/2020**

1 INTRODUCTION

Methods presented in this chapter define how to calculate the stress and strain distribution in the section of a beam made of composite materials, i.e. a section composed of segments of fibrous composite laminates.

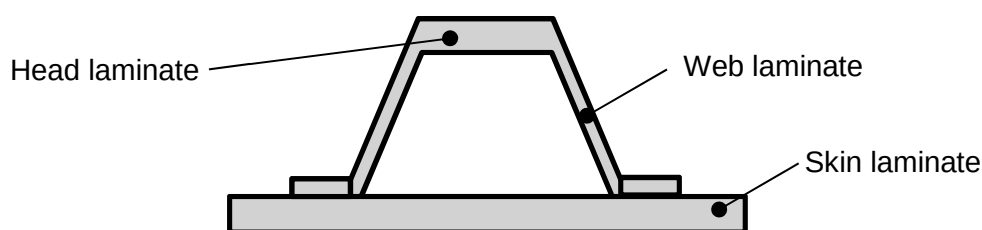


Figure 1. Cross section of a hat/omega stiffener bonded to a skin, made of composite materials

1.1 METHOD-SUMMARY

The method is essentially an extension of the classical beams theory presented in chapter §3110 for isotropic alloys, but adapted to composites. Basic assumption is still that cross-sections normal to the beam axis remain plane and normal to that axis. This assumption, combined with the classical laminate theory (see chapter §3401) allows to derive formulation equivalent to that for isotropic materials.

1.2 SCOPE AND LIMITATIONS

Stress analysis of straight beams of composite materials.

Similar limitations as for beams made of isotropic materials (see chapter §3110).

2 METHOD DESCRIPTION

2.1 Strains and stresses under Axial Loads and Bending

As mentioned previously, the method is essentially an extension of the classical beams theory presented in chapter §3110 for isotropic alloys, but adapted to composites.

Equilibrium equations

At each section, if X is aligned with the beam axis (axial direction), Y/Z represent the local reference directions within the cross section, the relationships between axial load P_x , and bending moments M_y , M_z , with the axial stress distribution and cross-section geometry can be expressed through integrals across the section of area A as:

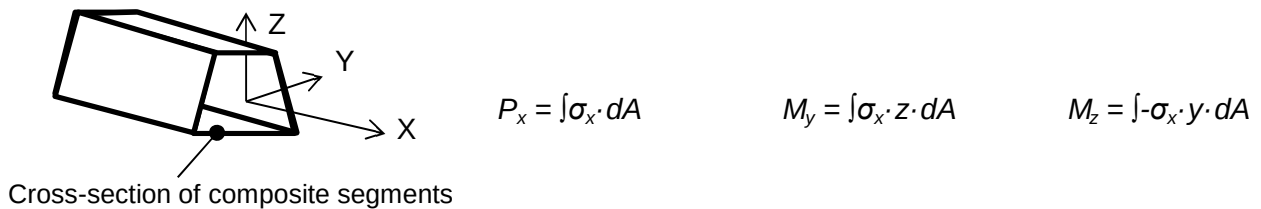


Figure 2. Loads equilibrium in a composite beam section

In absence of transversal axial stresses ($\sigma_y = 0$), for laminates following standard design rules (with no shear-extensional / membrane-bending couplings), in each laminate segment the relationship between stress and strain can be expressed as $\epsilon_x = \sigma_x / E_x$, leading to the following equations:

$$P_x = \int \epsilon_x \cdot E_x \cdot t \cdot ds \quad M_y = \int \epsilon_x \cdot E_x \cdot t \cdot z \cdot ds \quad M_z = \int -\epsilon_x \cdot E_x \cdot t \cdot y \cdot ds$$

The solution for the strain distribution has expressions similar to those for the stress in isotropic elastic beams...

Tension and bending in composite materials

In pure axial loads the solution is straightforward (uniform strain):

$$P_x = \epsilon_x \cdot \int E_x \cdot t \cdot ds = \epsilon_x \cdot \sum (E_x \cdot b \cdot t)_i = EA \cdot \epsilon_x \quad , \text{ or } \quad \epsilon_x = P_x / EA \quad , \text{ with } \quad EA = \sum (E_x \cdot b \cdot t)_i$$

Where as usual, for each laminate-segment of the section: $E_x = 1 / (a_{11} \cdot t)$

...which, for balanced laminates, is equal to $E_x = (A_{11} - A_{12}^2 / A_{22}) / t$

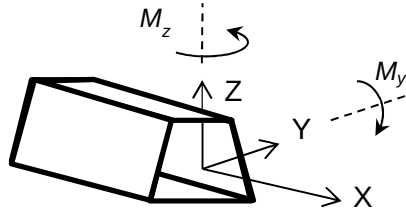
This solution is exact if no extension-bending coupling is present, which is the case for laminates with $[B] = 0$ (e.g. symmetric laminates) or if loads are related to the neutral plane of the laminate instead of to the mid-plane. In case the distance between neutral and mid-plane with regards to the section dimensions is very small, which is the usual case, this assumption is acceptable and the expressions in this chapter are valid.

For linear elastic materials in bending, the neutral axis always passes through a reference point at the cross section plane (in beams made of isotropic materials, this point is the centre of gravity). In the

case of composites sections, this reference point must be calculated weighting the areas of the flat flanges/segments composing the section with their elastic moduli:

$$(y, z)_{CG} = \int E_x \cdot (y, z) \cdot dA / \int E_x \cdot dA = \sum \{ E_x \cdot b \cdot t \cdot (y, z) \} / \sum (E_x \cdot b \cdot t)$$

Using a local reference axis with origin in that reference point of the section, and with axes matching the principal axis system (simple in sections with any axis of symmetry), the strain distribution across the section in combined axial and bending loads becomes:



$$\epsilon_x = P_x / EA + M_y \cdot z / EI_y - M_z \cdot y / EI_z$$

...where:

$$EA = \int E_x \cdot dA \quad , \quad EI_y = \int z^2 \cdot E_x \cdot dA \quad , \quad EI_z = \int y^2 \cdot E_x \cdot dA$$

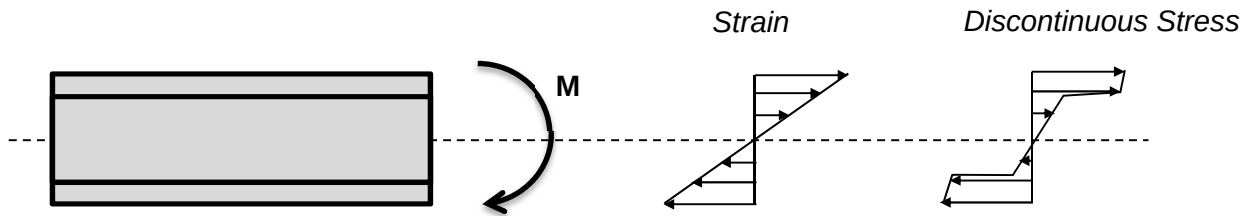


Figure 3. Bending strain and stress in a composite beam

Section properties EI_y and EI_z define its bending stiffness. Typically, abovementioned reference point and the bending inertias are calculated by concentrating, on each segment, the thickness weighted with the laminates apparent Young's modulus at the mid-line. The contributions of the $[D]$ matrices to the section bending stiffness are sometimes neglected but can be computed.

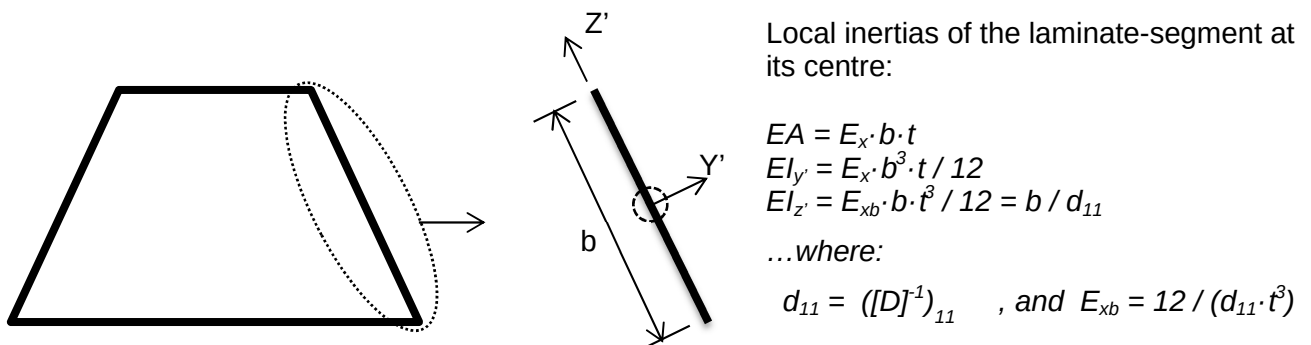
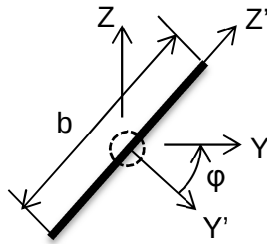


Figure 4. Decomposition of a composite section in segments

Steiner's theorem and transformation rules of bending moments of inertia in isotropic materials can be applied also to its E_x -weighted pairs in composite sections.



$$\begin{aligned}\Pi &= (El_{y'} + El_{z'}) / 2 \\ \Delta &= (El_{y'} - El_{z'}) / 2 \\ El_y &= \Pi + \Delta \cos(2\phi) - El_{yz'} \sin(2\phi) \\ El_z &= \Pi - \Delta \cos(2\phi) + El_{yz'} \sin(2\phi) \\ El_{yz} &= \Delta \sin(2\phi) + El_{yz'} \cos(2\phi)\end{aligned}$$

Figure 5. Transformation of Moments of Inertia due to axes rotation

For quasi-orthotropic laminates ($[B] = 0$), the product of inertia $El_{yz'}$ in local laminate/segment reference axis system shall be typically neglected and expressions above simplified. Additionally, in sections with some symmetry axis, the resultant product of inertia El_{yz} of the whole section is zero.

2.2 Shear and Torsion in thin-walled sections

Shear stresses in thin-walled sections can be calculated thanks to the classical theory of elasticity and Collignon's theorem (shear strains are uniform across the thickness), as for sections made of isotropic materials, using the yet familiar transformations with the apparent elastic moduli and stiffness matrices of the laminate segments. Formulas in chapter §3110, sub-section 2.2, would become:

Detail	Load	Isotropic	Composite
Thin-wall OPEN	Vertical Shear V_z	$\tau = V_z \cdot Q_y / (I_y \cdot t)$	$N_s = G \cdot t \cdot \gamma = V_z \cdot EQ_y / El_y$
Thin-wall OPEN	Torsion T	$\tau' = T \cdot t / J$ $J = \Sigma(b \cdot t^3 / 3)_i$	$\gamma' = T \cdot t / GJ$ $GJ = 4 \cdot \Sigma [b \cdot (D_{66} - D_{26}^2 / D_{22})]_i$
Thin-wall CLOSED	Vertical Shear V_z	As explained in §3110-2.2: split the section at a segment, add a constant shear flow as unknown, superimpose those stress distributions, and state the forces equilibrium	
Thin-wall CLOSED (enclosed area A)	Torsion T	$\tau = T / (2 \cdot A \cdot t)$	$N_s = G \cdot t \cdot \gamma = T / (2 \cdot A)$

Table 1. Transformation of shear equations in composite beam sections

Additional notation introduced in previous table:

- N_s shear load within the laminate plate
- τ shear stress (the apparent shear stress in composite laminates),
- γ engineering shear strain
- τ' shear stress at the most outer faces in plate torsion

- γ' shear strain at the most outer faces in plate torsion
- G shear modulus of the laminate
- GJ torsion stiffness of the laminate section of width b
- Q first moment of area at the point of interest

Other remarks:

- For calculating the bending moment of inertia EI_y and the first moment of area EQ_y , the integrals of area were weighted with the elastic modulus for each laminate/segment
- Through thickness laminate stresses are not treated in this chapter revision
- Halpin presents the torsion stiffness of a laminated beam as $GJ / b = 4 / [D^{-1}]_{66}$ (see Ref. 3, sub-section 5.8). The expression in the table account for laminates with non-zero twisting-bending coupling terms, derived by authors other authors as Weisshaar (Ref. 4).

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A	mm ²	Area of the cross section
[A]	N/mm	In-plane stiffness matrix of the laminate
A _{ij}	N/mm	Term (i,j) of the in-plane stiffness matrix [A], i.e. using typical array notation: $A_{11} = A(1,1)$, $A_{12} = A(1,2)$, $A_{22} = A(2,2)$, $A_{66} = A(3,3)$
a ₁₁	mm/N	Term (1,1) of the inverse of the in-plane stiffness matrix of the laminate [A] ⁻¹
b	mm	Width of a flat flange/segment in a composite beam section
[B]	N	Matrix defining the laminate shear-bending coupling of the laminate
[D]	Nmm	Bending stiffness matrix of a composite laminate
D _{ij}	N/mm	Term (i,j) of the bending stiffness matrix [D] , i.e. using typical array notation: $D_{11} = D(1,1)$, $D_{12} = D(1,2)$, $D_{22} = D(2,2)$, $D_{16} = D(1,3)$, $D_{26} = D(2,3)$, $D_{66} = D(3,3)$
d ₁₁	(Nmm) ⁻¹	Term (1,1) of the inverse of the bending stiffness matrix of the laminate [D] ⁻¹
EA	N	Axial stiffness of the laminated beam
EI	Nmm ²	Bending stiffness of the laminated beam
E _x	MPa	Apparent elastic modulus of the laminate in X direction, equal to $1/(a_{11} \cdot t)$
E _{xb}	MPa	Apparent bending modulus of the laminate in X direction, equal to $12/(d_{11} \cdot t^3)$
G	MPa	In-plane shear modulus of the laminate
GJ	Nmm ²	Torsion stiffness of the laminated beam
J	mm ⁴	Torsional constant/inertia of the cross-section
M	Nmm	Bending moment
N _s	N/mm	In-plane shear load in the laminate
Q	mm ³	First moment of area of a section at a point
P	N	Axial force
t	mm	Thickness of the laminate
T	Nmm	Torsion moment
γ	-	Engineering in-plane shear strain in the laminate
τ	MPa	Shear stress

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 2	[Christos Kassapoglou]	2 nd Ed.	Design and Analysis of Composite Structures
Ref. 3	[John C. Halpin]	2 nd Ed.	PRIMER ON COMPOSITE MATERIALS ANALYSIS
Ref. 4	[T.A. Weisshaar, B.L. Foist]	1985	Vibration Tailoring of Advanced Composite Lifting Surfaces

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 2. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 06/03/2020	G. Baños	First issue	All pages

Table 3. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 06/03/2020	Florian Glock 06/03/2020

Table 4. Approval signatures table for last revision of this chapter